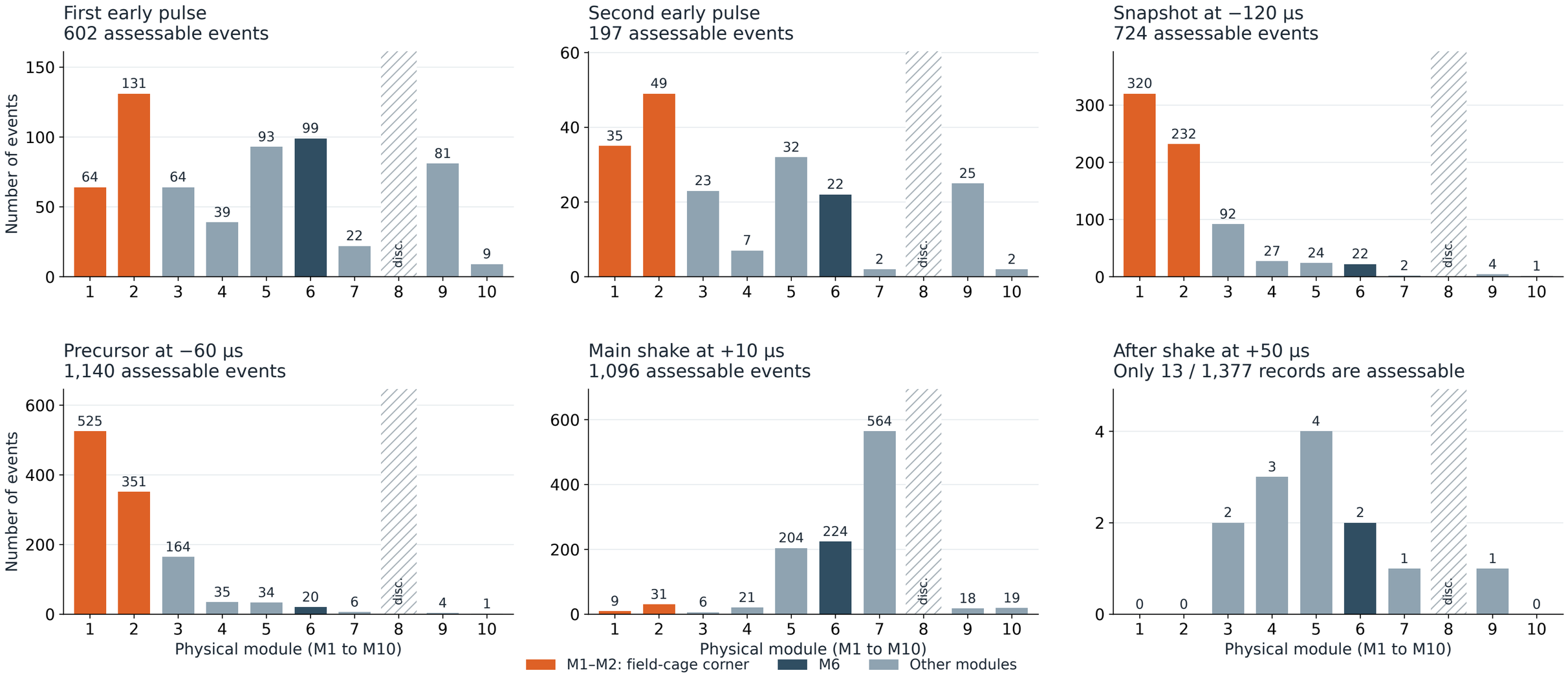
Where is the largest recorded PDS response?

All 1,377 available APA2 records with a main-shake time · One entry is one event with a clear largest module.



Taller bars mean more events. This measures the largest response, not the earliest onset or the amount of light.

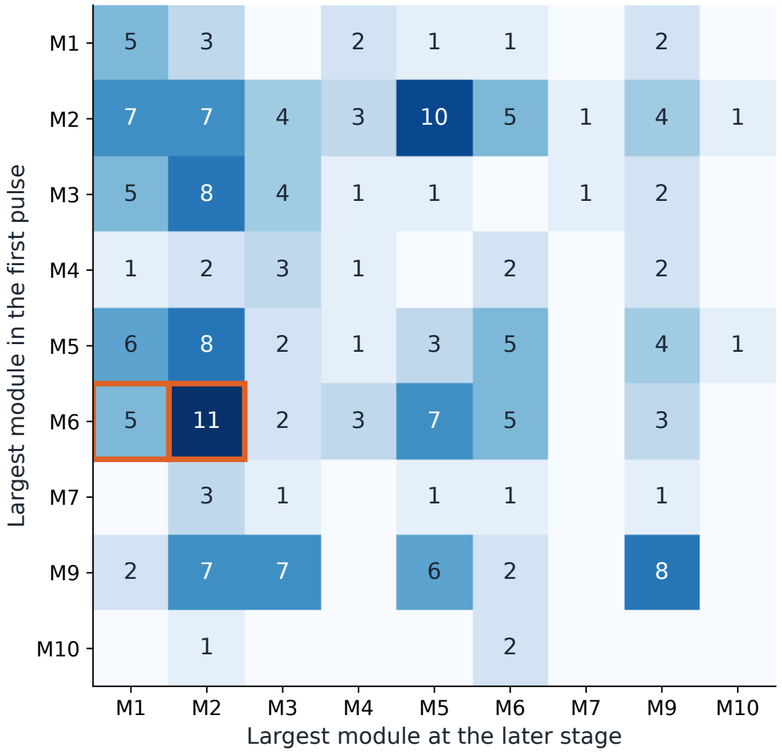
Each stage can use different events. At least 12 recorded connected inputs across 4 modules; largest signal ≥20 ADC counts. Tied modules are omitted.

Early pulses: first two existing sharp candidates before −100 µs. Snapshots span 0.512 µs. M8 is reported disconnected; no missing sample is treated as zero.

Does an event change from M6 to the corner?

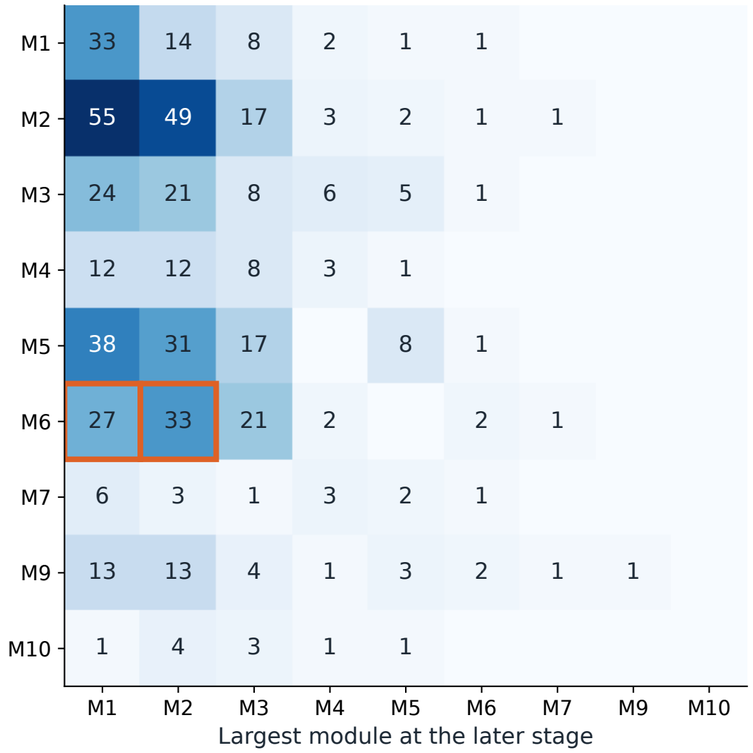
Each square counts events. Both stages use exactly the same recorded connected channels within each event.

First pulse → second early pulse
197 paired events



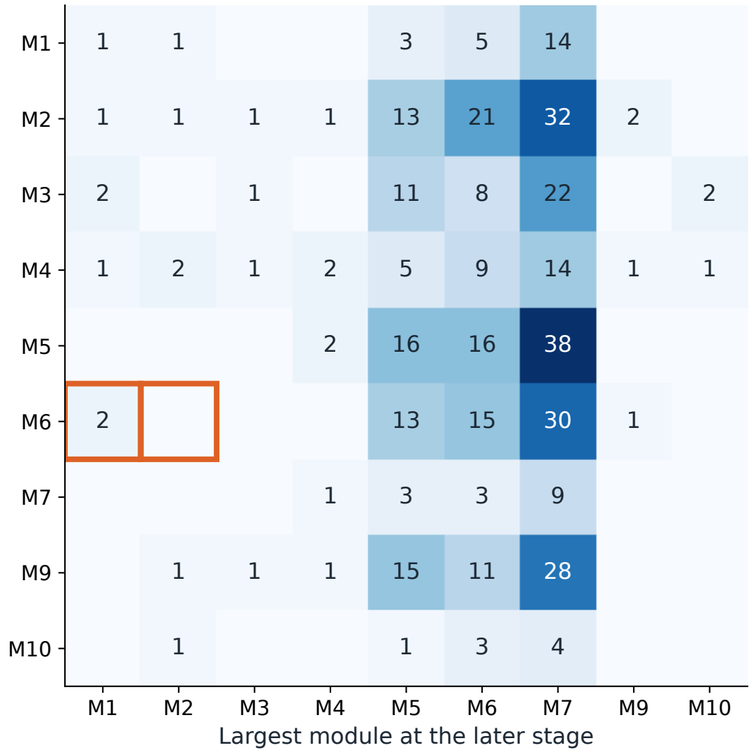
First pulse strongest in M6: 36 events
Later strongest in M1–M2: 16 / 36 (44%)

First pulse → precursor at −60 µs
533 paired events



First pulse strongest in M6: 86 events
Later strongest in M1–M2: 60 / 86 (70%)

First pulse → main shake at +10 µs
393 paired events



First pulse strongest in M6: 61 events
Later strongest in M1–M2: 2 / 61 (3%)

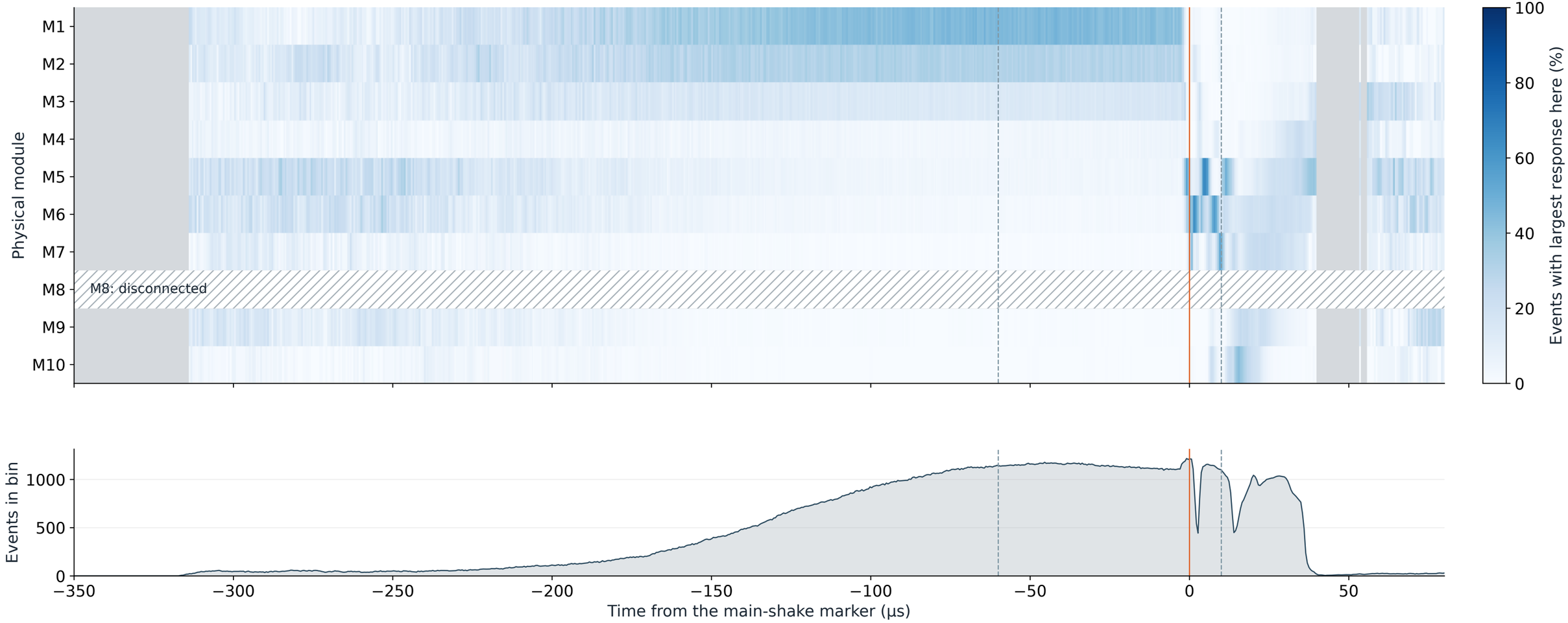
Orange boxes are the proposed M6 → M1/M2 change. The strongest section does change in some events; it does not follow one fixed route.

Each pair requires at least 12 common connected inputs across 4 modules and a signal ≥20 ADC counts at both stages.

This is a change in recorded response. Channel gains, signal polarity and electronics can affect it; no moving source or propagation speed is fitted.

The strongest section changes with time across the sample

At each time: percentage of assessable events whose largest recorded signal is in that module. The colour scale is fixed.



Grey means fewer than 20 assessable events in that time bin. The lower plot shows the denominator; it changes with recording coverage.

Early pulses occur at different times and spread out in this view. Use the paired plots to check a change within the same event.

Each bin spans 0.512 µs. These are event counts, not summed ADC amplitudes. Missing and rail-flagged samples are excluded.

What supports the M6-to-corner observation?

The shift toward M1–M2 during the slow precursor remains in both November runs and in the comparison checks.

Comparison	Paired events	M6 first	Then M1–M2	Share of M6-first
Run 32731	320	59	43 / 59	73%
Run 32739	213	27	17 / 27	63%
Both runs + one July record	533	86	60 / 86	70%
Exclude events 33 and 52	531	85	59 / 85	69%
Use a 4.096 µs window	544	93	73 / 93	78%
Raise the signal cut to 50 ADC	428	70	53 / 70	76%
Exclude header-only baselines	462	76	55 / 76	72%
Require pulse before TPC precursor	355	69	47 / 69	68%

Coverage of the saved sample

1,531 saved records were checked; 1,527 contain PDS data.

1,377 have both APA2 endpoint 109 and a main-shake time: 714 in run 32731, 662 in run 32739, and one in run 27938.

15 records have no main-shake time. Another 139 aligned records have no APA2 endpoint 109.

The October files contain other PDS endpoints. Applying the APA2 M6/corner layout to them would give the wrong physical comparison.

After the shake, only 13 records pass the snapshot coverage rule; only 4 support a main-to-after paired comparison.

The counts describe this saved sample, not the frequency of this behaviour during detector operation.

M6 has the largest first pulse in 99 / 602 assessable events. The M6-to-corner sequence recurs, but M6 is not the usual starting maximum.

“Largest” is an amplitude comparison. It is not the same as “first to start”; channel timing is checked separately in the linked leading-channel study.